Python Asset Inventory Tool

A command-line tool for recording and managing IT assets including devices, operating systems, assigned users, and asset status in a structured CSV inventory.

Features

- ✓ **Add Assets** - Record new devices with unique auto-generated IDs
- ✓ **View Inventory** - Browse all assets in your inventory
- ✓ **Search** - Find assets by hostname, user, OS, department, or any keyword
- ✓ **Update Status** - Change asset status (Active, In Storage, Retired, Repair)
- ✓ **Export Reports** - Generate summary reports with statistics
- ✓ **CSV Storage** - Easy to edit in Excel or Google Sheets
- ✓ **No Dependencies** - Uses only Python standard library

Requirements

- Python 3.6 or higher
- No external packages required

Quick Start

1. Run the Tool

```
python asset_inventory_tool.py
```

2. Use the Menu

The tool will display an interactive menu:

```
================================================
MAIN MENU
================================================
1. Add New Asset
2. View All Assets
3. Search Assets
4. Update Asset Status
5. Export Report
6. Exit
================================================
```

3. Add Your First Asset

Select option `1` and enter asset details:

```
Device Type* (e.g., Laptop, Desktop, Server, Printer): Laptop
Hostname* (e.g., WIN-LP-001): WIN-LP-001
Operating System* (e.g., Windows 11, Ubuntu 22.04): Windows 11 Pro
Assigned User* (e.g., John Dlamini): John Dlamini
Department (e.g., IT, HR, Finance): IT
Location (e.g., Big Bend Office, Remote): Big Bend Office
Purchase Date (YYYY-MM-DD): 2024-01-15
Warranty Expiry (YYYY-MM-DD): 2027-01-15
Status* (Active, In Storage, Retired, Repair): Active
Notes (optional): Dell Latitude 5520
```

Data Fields

Field	Required	Description
asset_id	Auto	Unique identifier (AST-YYYYMMDDHHMMSS)
device_type	Yes	Laptop, Desktop, Server, Printer, etc.
hostname	Yes	Network hostname (e.g., WIN-LP-001)
operating_system	Yes	OS name and version
assigned_user	Yes	Person responsible for the asset
department	No	Department or team
location	No	Physical or remote location
purchase_date	No	Date of purchase (YYYY-MM-DD)
warranty_expiry	No	Warranty expiration date
status	Yes	Active, In Storage, Retired, Repair
notes	No	Additional information
last_updated	Auto	Timestamp of last modification

Example Use Cases

Track IT Equipment

Record all company laptops, desktops, and servers with their assigned users and locations.

Monitor Warranty Status

Track purchase dates and warranty expiry dates for budget planning.

Audit Compliance

Export reports for security audits or insurance documentation.

Remote Work Inventory

Track which devices are in the office vs. remote locations.

File Structure

The tool creates two files:

- **asset_inventory.csv** - Main inventory database (CSV format)
- **asset_report_YYYYMMDD_HHMMSS.txt** - Exported reports

CSV Format

The inventory is stored in standard CSV format, compatible with:

- Microsoft Excel
- Google Sheets
- LibreOffice Calc
- Any text editor

Automation Examples

Bulk Import from Another System

You can programmatically add assets by importing the CSV module:

```
import csv
from asset_inventory_tool import add_asset, initialize_inventory

initialize_inventory()

# Add multiple assets
assets = [
    ("Laptop", "WIN-LP-010", "Windows 11", "User1", "IT", "Office"),
    ("Desktop", "LIN-DT-011", "Ubuntu 22.04", "User2", "Dev", "Remote"),
]

for device, hostname, os, user, dept, loc in assets:
    asset_id = add_asset(device, hostname, os, user, dept, loc, "", "", "Active")
    print(f"Added: {asset_id}")
```

Integrate with Active Directory

For advanced users, you can extend the tool to pull user information from Active Directory using the `ldap3` or `pyad` libraries.

Scheduled Reports

Use a cron job (Linux) or Task Scheduler (Windows) to export reports automatically:

```
# Linux cron example - daily at 6 AM
0 6 * * * cd /path/to/tool && python -c "from asset_inventory_tool import export_repo
```

Troubleshooting

"No inventory file found"

- Add at least one asset first, or the file will be created automatically on first use.

"Permission denied"

- Ensure you have write permissions in the directory where the script is located.

CSV file corrupted

- The CSV format is simple. You can manually edit it in a text editor if needed.
- Always keep a backup before manual edits.

Customization

Change Default File Location

Edit the `CSV_FILE` variable at the top of the script:

```
CSV_FILE = "/path/to/your/inventory.csv"
```

Add Custom Fields

Add new field names to the `FIELDNAMES` list:

```
FIELDNAMES = [
    "asset_id",
    "device_type",
    # Add your custom fields here
    "serial_number",
    "vendor",
    # ... rest of fields
]
```

Security Notes

- The CSV file is stored in plain text - protect it with appropriate file permissions
- For sensitive environments, consider encrypting the file or using a database backend
- Regular backups are recommended

License

This tool is provided as-is for educational and practical use. Feel free to modify and extend for your needs.

Support

For issues or feature requests, review the code comments or extend the functionality based on your requirements.

Created: 2026
Python Version: 3.6+
Dependencies: None (standard library only)